



Tzchen (Edison) Wu

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Education

University of Nevada – Las Vegas

December 2027

Bachelor of Science in Electrical Engineering, Minor in Japanese

Technical Skills

UC OASIS Foundations – Semiconductor Processing & IC Fabrication

Apr 2026

Programming: Swift, SystemVerilog, C, C++

Software & Tools: Arduino, Git/GitHub, Docker, Linux/Ubuntu, Quartus II, LTspice, MATLAB, SolidWorks

Hardware: FPGA, circuit prototyping, breadboarding, soldering, 3D printing

Languages: English (Native), Mandarin (Native), Japanese (Conversational)

Projects

Eddie Calendar – iOS Calendar Application

May 2026 – Present

- Developing and daily driving a native Swift iOS calendar integrating **3 event sources** – Google Calendar, Apple Calendar, and locally created events
- Built 4 custom calendar views with filtering and customizable visibility for faster schedule access
- Integrated external data from Apple WeatherKit and a news API to generate daily look-ahead notifications

Personal AI Location Server – Ubuntu, Docker, Dawarich, Tailscale, Ollama

March 2026

- Deployed a self-hosted, containerized server to ingest, store, and visualize iPhone GPS data at **30-second intervals**
- Integrated Ollama for local natural-language querying, summarization, and analysis of location history
- **Configured a Tailscale VPN for secure remote data transmission and mobile access**
- Troubleshoot networking, containers, database ingestion, GPU acceleration, and mobile-to-server integration

USB Power Supply & Charging Circuit

April 2026

- Designed and prototyped a regulated 5V USB power supply using an LM7805, validated the circuit in **LTspice** before **breadboard** testing and final **perfboard soldering**.
- Verified voltage and current behavior on both the breadboard prototype and soldered perfboard using a **Keithley 2110 digital multimeter**
- Implemented USB charging signaling and verified real-device operation by charging iPhones and other USB devices

Car Turn Signal Controller – SystemVerilog, Quartus II, FPGA

September 2024

- Designed a **7-state Moore FSM-based** turn-signal controller for a 1965 Ford Thunderbird in **SystemVerilog**
- Simulated and verified FSM operation and output waveforms using **ModelSim**
- Implemented the design in **Quartus II** on a **DE2-115 FPGA** and verified operation on physical hardware

Work Experience

UGym Technology Corp. – Sales & Marketing Intern, CES

Jan 2023 & Jan 2024

- Represented UGym Technology at **2 CES exhibitions**, explaining products to visitors and providing real-time Mandarin-English interpretation
- Demonstrated products, answered questions, and spoke directly with prospective customers

Surglasses – Engineering Intern

Feb 2022 – Mar 2022

- **Soldered wires and connectors**, applied **heat-shrink insulation**, and followed schematics to **assemble electronic wiring**
- Revised the English-language user manual for an electronic product originally written in Mandarin

Point Robotics Medtech Inc. – Engineering Intern

Jun 2021

- Modeled parts in **SolidWorks**, prepared them for **3D printing**, and gained hands-on experience with the prototyping process
- Translated and revised a company press release from Mandarin to English

Leadership

Founder & President, UNLV Run Club

Aug 2026 – Present

- Completed the documentation and registration process to establish an official **RSO at UNLV**
- Designed and distributed **75 flyers** around campus to promote the club and build awareness
- Organize, market, and host **weekly group runs** for club members

UNLV Marching Band – Trombone Section Leader

Aug 2024 – May 2026

- Coordinated movement and logistics for **19 trombones, including 9 freshmen**, and split the section into **2 balanced groups** as needed
- Led sectionals, tracked attendance, conducted uniform checks, and communicated rehearsal expectations and updates to the section